MILITARY SPECIFICATION

CANNON, 175MM GUN: M113A1

This specification is approved for use within the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification supports the acquisition of Cannon, 175MM, H113A1 for use with the Gun, Field Artillery, S.P.: 175MM, H107. The cannon consists of a tube and a breech mechanism assembly. This specification includes the minimum essential Engineering and Packaging requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifciations and standards form a part of this specification to the extend specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SAS, Watervliet, NY 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1025

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SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specifications for

MIL-P-14232 - Parts, Equipment, and Tools for Army Material, Packaging of

MIL-I-45607 - Inspection Equipment, Acqisition, Maintenance, and Disposition of

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1189 - Standard DOD Bar Code Symbology

MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

MIL-STD-2073-1 - DOD Material Procedures for Development and Application of Packaging Requirements

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U. S. Army Armament. Munitions and Chemical Command

EPL 11578119 - Engineering Parts List for Cannon, 175MM Gun: M113A1

LQAP 11578119 - List of Quality Assurance Provisions for Cannon. 175MM Gun: M113A1

018767541 - List of Inspection Equipment for Breed Assembly Mechanism Cannon, 175MM Gun: M113A1

DL 11578119 - List of Inspection Equipment for Cannon, 175 MM Gun: M113A1

PDS P11578119 - Packaging Data Sheet for Cannon, 175MM Gun: H113A1

PUBLICATIONS

QAP-APPENDIX-WVA - General Quality Assurance Provisions

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(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

2.2 Other publications. The following documents form a Part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM D3951 - Commercial Packaging, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents may also be available from libraries or other information services).

2.3 Order of precedence. The applicable contract contains an order of precedence clause wherein the specifications are listed last. With respect to this specification, the order of precedence is drawing requirements, followed by the text of this specification, then the other referenced specifications and standards. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General Requirements. Cannon and component parts shall be in accordance with the applicable contract, the applicable drawing [EPL 11578119), the requirements of this specification, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article uless it is specifically waived in the contract (see 4.3 and 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer (see 6.3).

3.3 Functioning All parts shall function smoothly without interference, erratic movement, or malfunction.

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3.3.1 Breech mechanism (Opening). All parts of the breech mechanism shall operate without interference, erratic movement, or malfunction during opening and closing operations. For opening, a rearward and downward pull on the operating-handle, released from its latched position, shall rotate the breechblock counterclockwise to disengage the sectored threads. A push to the right on the operating-handle shall cause the breechblock to swing out of the breech to the open position. The breech mechanism shall be retained in the open position by the counterbalance.

3.3.2 Breech mechanism (Closing). In the open position, a pull on the operating-handle to the left past the dead-center position of the hinge-pin arm shall swing the breechblock into the breech opening. This action allows the roller on the breechblock to strike the cam and rotate the breechblock clockwise to engage the sectored threads. An upward push on the operating-handle shall complete the rotation of the breechblock and lock the breech. The operating-handle shall latch in the closed position.

3.3.3 Obturator surface. The obturator surface shall conform to the requirements prescribed in MIL-C-13931.

3.3.4 Block assembly. When the breech mechanism is moved to the open position, the block assembly, with a percussion type firing mechanism mounted in position, shall slide vertically downward into the extract position. The extractor shall then forcibly eject the primer cartridge out of the spindle chamber. When the breech mechanism is moved to close the breech, the block assembly shall be automatically cammed from the extract position to the load position. During the

chamber. Only when the breechblock is closed and locked shall the block assembly be capable of moving to the closed position and of being moved manually into the load and extract positions.

3.3.5 Firing pin protrusion and retraction. The firing pin protrusion, measured from the front face of the block, shall not be less than 0.114 inch and not more than 0.118 inch in the fired position. Firing pin-retraction behind the front of the block shall not be less than 0.012 inch.

3.3.6 Primer headspace. Primer headspace, the distance between the front face of the block assembly and the rear face of the obturator-spindle, shall not be less than 0.052 inch nor more than 0.056 inch.

3.3.7 Functional location of recoil cylinder rod hole in the breech ring to rails. The functional location of recoil cylinder rod hole in the breech ring, in relation to the rails on the assembled cannon, shall be functionally checked with inspection equipment designed from dimensions, either explicit, derived, or calculated, and shall conform to applicable drawings listed in EPL 11578119.

3.4 Performance.

3.4.1 High pressure resistance. The cannon and repair tubes shall be capable of withstanding a chamber pressure of 54,700 + 1900 psi when tested as specified in 4.4.5.

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3.4.2 Material soundness. After completion of initial production or accetpance testing, all parts shall be free from cracks and fractures and shall conform to the soundness requirements specified on the applicable drawings.

3.4.3 Bore enlargement. Bore diameters, measured across the rifled lands after proof firing no more than 8 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amount specified in Table 1.

TABLE I

Distance forward of Commencement of rifling	0.1-Inch	l-Inch	I-Cal.	2-Cal . to muzzle
Amount (in inch)	0.009	0.009	0.008	0.006

NOTE: Interpolate linearly to determine the allowable bore enlargement at distance other than-the distances given.

3.5 Workmanship Workmanship shall be in accordance with the requirements of MIL-C-13931

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable. for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it comnit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. A first article shall be submitted for inspection in accordance with contract requirements. The fist article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the inspections, examinations, and tests specified herein and such other inspections, examinations, and tests as are necessary to determine compliance with the requirements of the contract.

4.4 Quality conformance inspection.

4.4.1 Inspection provisions. Cannon shall be inspected for conformance to requirements of this specification, MIL-C-13931, QAPs listed in Master LQAP 11578119, and QAP-APPENDIX-WVA.

4.4.1.1 Final examination. Final examination of each cannon shall be completed after testing and prior to packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract. The outer surfaces of each finish machined tube, the inner and outer surfaces of each breechblock, breech ring and bushing shall be magnetic particle tested in accordance with the applicable part drawing. The inner surface of the tube shall be examined by means of a borescope. Any evidence of cracks or fractures shall be cause for rejecttion.

4.4.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.4.2 Inspection equipment. Data lists containing drawing numbers of Government-designed inspection equipment used in the inspection of parts, subassemblies, and assemblies are listed in DL 11578119 and DL 8767541.

4.4.2.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2.1), responsibility for acquistition, maintenance, and disposition of acceptance inspection and test equipment precribed on the List *Of* Inspection Equipment DL 11578119, DL 8767541 and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.4.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.4.3 Examinations.

4.4.3.1 Firing pin protrusion and retraction. Firing pin protrusion and retraction shall be gaged on each cannon with the applicable gages referenced on the List of Inspection Equipment, DL 8767541, or with approved equivalent gages. Cannon failing to meet the requirements of 3.3.5 shall be subject to rejection.

4.4.3.2 Primer headspace. Primer headspace shall be gaged on each cannon with the applicable gage referenced on List of Inspection Equipment, DL 8767541, or with an approved equivalent gage. Cannon failing to meet the requirements of 3.3.6 shall be subject to rejection.

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(4.4.3.3 Functional location of *recoil* cylinder rod hole in the breech ring to rails. Each assembled cannon shall be inspected for functional location of recoil cylinder rod hole in the breech ring to rails using the applicable gage (D8772085) referenced on List of Inspection Equipment, DL 8767541, or with an approved equivalent gage. Cannon failing to meet the requirements of 3.3.7 shall be subject to rejection.

4.4.3.4 Workmanship. Workmanship shall be examined visually in accordance with MIL-C-13931.

4.4.4 Tests.

4.4.4.1 Functioning tests. Functioning tests shall be conducted in accordance with MIL-C-13931 and as specified herein. A minimum of two complete cycles of operation shall be performed on each cannon.

4.4.4.1.1 Breech mechanism. Each breech mechanism shall be operated manually to determine whether the functional requirements of 3.3.1 and 3.3.2 are met. The action of the operating-handle shall be observed to assure that the handle will latch in the closed position after each complete cycle of operation. Any evidence of interference, erratic movement, or malfunction of any part shall be cause for rejection.

4.4.4.1.2 Block assembly. Each block assembly with the firing mechanism mounted in position, shall be operated manually to determine whether the functional requirements of 3.3.4 are met. Any evidence of interference, erratic movement, or malfunction of any part shall be cause for rejection.

4.4.4.1.3 Obturator. The obturator surfaces shall be examined and the bearing or obturator parts shall be tested in accordance with MIL-C-13931. Failure to comply with the requirements of 3.3.3 shall be cause for rejection.

4.4.5 Proof acceptance tests. Cannon and repair tubes shall be proof fired at a Government specified proving ground to assure compliance with 3.4. The following procedures apply:

a. Each cannon or repair tube shall be proof fired once at 54,70031900 psi.

b. A facility mount may be used if necessary.

c. Chamber pressure shall be measured by placing pressure gauges on either side of the charge, one at the 10 o'clock and one at the 2 o'clock position, as close to rear of the charge as possible. Gages shall not be placed below or behind the charge. All measurements shall be recorded as specified in 4.4.5.1.

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d. After proof firing:

(1) The bore of each tube shall remeasured for bore enlargement using an air gauge, an electronic gauge or other approved gauge. The measurements shall be recorded. Bore enlargement shall meet the requirements of 3.4.3 and Table I.

(2) Each tube shall be magnetic particle inspected and borescoped.

(3) Each breech mechanism shall be examined for looseness or interference of fit.

(4) Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks and fractures. Each breech ring, breech ring bushing and breechblock shall be magnetic particle inspected.

(5) Each breech mechanism shall be cleaned and reassembled. It shall then be functioned in accordance with 4.4.4.1.

e. Cannon or repair tubes found to be defective at the proving ground shall be reported to the Procuring Contracting Officer; the Product Assurance Directorate, Watervliet Arsenal, Matervliet, NY 12189-4050; and Benet Laboratories, Watervliet Arsenal, Matervliet, NY 12189-4050; within one working day. The test agency shall forward a Test Incident Report (TIR) of the incident within three working days.

4.4.5.1 Weapon record data. Each cannon and repair tube shall be furnished with applicable data on DA Form 2408-4 (Weapon Record Data), as specified by the Procuring Contracting Officer.

4.4.6 Packaging inspection. Unless otherwise specified, inspection to determine compliance with packaging requirements of the applicable Packaging Data Sheet (PDS) for the level designated in the contract shall be as specified in MIL-P-14232.

5. PACKAGING

5.1 Cannon - Levels A, B and C. Cleaning, drying, preservation, unit packaging and packing of cannon shall be in accordance with the requirements of PDS P11578119 and MIL-P-14232 for the level of protection specified in the contract (see 6.2).

5.2 Repair parts - Levels A, B and C. Cleaning, drying, preservation, unit packaging, and packing of repair parts shall be in accordance with repair part Special Packaging Instruction and MIL-STD-2073-1 (or the applicable repair part Packaging Data Sheet and MIL-P-14232) for the level of protection specified in the contract (see 6.2.).

5.2.1 Level X - Commercial packaging. drying, preservation, unit packaging and packing of repair parts shall be in accordance with ASTM D3951 when industrial packaging is specified in the contract (see 6.2).

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5.2.1.1 Quantity in Unit Package (W). When commercial packaging of repair parts is specified in the contract, the QUP shall be the same as specified in the contract (see 6.2.).

5.3 Marking of packages.

5.3.1 Levels A, B and C. Marking of unit packages, intermediate packages and exterior shipping containers shall be in accordance with MIL-STD-129. When specified (see 6.2), bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

5.3.2 Level X - Commercial. Marking of unit packages, intemediate packages and exterior shipping containers shall be in accordance with ASTM D3951. When specified (see 6.2), bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. The cannon specified herein is intended to be used in the 175-mn MI07 Self Propelled Field Artillery System.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.2),

c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

d. All inspection and test data that is required.

e. At whose expense a retest may be performed.

f Serial numbers for the cannon and repair tube assemblies.

g. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.4.2.1).

h. Availability of inspection equipment from the Government.

i. Extent of supplier's responsibility for Government furnished and for supplier-required final inspection equipment.

j. Applicable acceptance test procedures.

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k. Under what conditions simulated firing may be performed, then test equipment available, and the procedures to be followed.

l. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and repair parts (see 5.2).

m. When applicable, bar code marking requirements (see 5.3).

n. When warranted, the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

o. Unless otherwise specified, the application of MIL-I-45607 and MIL-STD-45662.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423. Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified-by an-approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.2	First Article	DI-T-4902

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L. , Volume 11, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Procuring Contracting Officer).

6.3 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Product Assurance Directorate at Watervliet Arsenal.

6.4 Subject term (key word) listing.

Cannon, 175-nmn: M113A1

Gun, Field Artillery, Self Propelled

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.6 International agreegment. All provisions of this specification are the subject of international standardization agreement, Q-STD-150. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Custodian:
Army -AR

Preparing activity:
Army-AR
Project No. 1025-A160

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-45524D	CANNON, 175MM GUN: M113A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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